

TAHDCO Unified Dashboard Portal — Software Testing Workbook

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Total Requirements 23	Total Test Cases 92	Passed 92	Failed 0	Blocked 0	Open Defects 0	Pass Rate 100.0%	Functional Tests 37	Functional Not Executed 0	Browser Checks 19
API Perf Tests 14	API Perf Passed 2	API Perf Not Executed 12	Module API Checks 15	Module API Passed 3	Module API Blocked 10				

1. Requirements (23 rows)

Req ID	Requirement	Description	Priority	Developer/Tester	Status	Remarks
REQ-01	Executive MD Dashboard	Unified Executive KPI Cards & District-wise Master Datable aggregating engineering & welfare metrics.	High	Frontend Dev	Implemented	Covered by DashboardMdComponent specs (UT-FE-01..06).
REQ-02	Engineering Tenders	TIPS Tender Management tracking work numbers, contractor allocations, financial values & milestone progress.	High	Full Stack Dev	Implemented	Covered by TenderComponent specs (UT-FE-30..35).
REQ-03	Housing Scheme	THMS Free Housing Construction Tracking with geotagged inspection photos & milestone level progress.	High	Full Stack Dev	Implemented	Covered by HousingComponent specs (UT-FE-36..41).
REQ-04	Attendance & Skill	TAMS Vocational Training & Trainee Enrollment Tracking across institutes & course completion rates.	Medium	Full Stack Dev	Implemented	Covered by EnrollmentComponent specs (UT-FE-42..47).
REQ-05	TAHDCO Welfare Schemes	District-wise Application Summary & Details for subsidy applications, DM & HQ approvals.	High	Full Stack Dev	Implemented	Covered by SchemeReportComponent specs (UT-FE-21..26).
REQ-06	CCTV Surveillance	Patrol360 High-Definition Site Camera Live Status, online/offline duration tracking & inspection feed.	High	Full Stack Dev	Implemented	Covered by Patrol360Component specs (UT-FE-48..53).
REQ-07	Welfare Board Cards	TNCWWB Construction Worker Member Registration & Welfare Claims tracking with Card Issued status.	High	Full Stack Dev	Implemented	Covered by TncwwbComponent specs (UT-FE-18..20).
REQ-08	Higher Education Loans	TELP Economic Loan Application Details tracking per district, division, and category type.	High	Full Stack Dev	Implemented	TELP data rendered inside the Scheme Report (REQ-05).
REQ-09	Official Diaries	TOD Task & Event Monitoring tracking overdue tasks, active events, and progress status.	Medium	Full Stack Dev	Implemented	Covered by TodComponent specs (UT-FE-54..59).
REQ-10	Scheduler Control	Dynamic Hangfire Job Control for automated API ingestion across all 7 project APIs (14 Endpoints).	Critical	Backend Dev	Implemented	Covered by ingestion/scheduler backend tests (UT-BE-01/07/08, UT-FE-69..71).
REQ-11	Preset Templates	1-Click Scheduler Job Preset Selector pre-populating Job Name, API URL, HTTP Method, and JSON Payload.	High	Backend Dev	Implemented	
REQ-12	RAG Vector Ingestion	Sub-second multi-project ingestion into vector search engine supporting LLM question answering.	High	Backend Dev	Implemented	Covered by GetRecordsAsync backend test (UT-BE-02).
REQ-13	1:1 Count Alignment	Enforce strict 1:1 match between clicked table cell counts and detail row lengths across all 7 projects.	Critical	Frontend Dev	Implemented	Covered by dashboard-md payload + drill specs (UT-FE-03..06, UT-FE-60..68).
REQ-14	Zero Filter Loss	Preserve District, Division, Year, Status, and Category filters seamlessly when navigating from Dashboard to Applications.	Critical	Frontend Dev	Implemented	Covered by TncwwbComponent queryParams restore spec (UT-FE-19).
REQ-15	Role Access Control	Role-based privilege enforcement restricting module access based on user role (Admin, MD, GM, CE, EE, DM).	High	Backend Dev	Implemented	Enforced on API (JWT apps claim) and UI (route guard). Covered by overview/shell specs (UT-FE-27..29, 72..78).
REQ-16	GIS Map Visualizer	Interactive Tamil Nadu District Map rendering live project markers, status badges, and hover tooltips.	Medium	Frontend Dev	Implemented	Rendered inside Executive MD Dashboard.
REQ-17	Global Header Search	Live Search dropdown matching works, beneficiaries, tenders, and district scopes in real time.	Medium	Frontend Dev	Implemented	

Req ID	Requirement	Description	Priority	Developer/Tester	Status	Remarks
REQ-18	PDF/Excel Export	One-click export of detailed report tables to CSV/Excel format for offline administrative analysis.	Medium	Backend Dev	Implemented	PDF via QuestPDF.
REQ-19	System Notifications	Centralized notification overlay modal filtering system alerts by project, frequency, and status.	Low	Frontend Dev	Implemented	
REQ-20	AI Copilot Widget	Floating AI Assistant widget for natural language querying of state-wide TAHDCO project metrics.	Medium	Full Stack Dev	Implemented	RAG-backed (REQ-12).
REQ-21	User Authentication	JWT-based login for registered users with role & privilege claims embedded in the token for session authorization.	Critical	Backend Dev	Implemented	Seeded demo users use password 'password123'. Covered by AuthService tests + login specs (UT-BE-04..06, UT-FE-10..14).
REQ-22	Audit Logging	Centralized audit trail recording Authentication, Ingestion, and Scheduler actions for admin review and filtering.	Medium	Full Stack Dev	Implemented	Covered by AuditLogComponent specs (UT-FE-15..17).
REQ-23	User Master	Full user CRUD with district assignment and a per-project privilege matrix managed from the User Master screen.	High	Full Stack Dev	Implemented	Covered by UserMasterComponent specs (UT-FE-79..83).

2. Unit Test Cases (92 rows)

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-BE-01	REQ-10	UnifiedIngestionService (BAL)	SyncAllProjectApisAsync_CompletesSuccessfully_And_ReturnsResult Verify a full sync across all project APIs completes and returns a batch result	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Mock HttpClient SendAsync to return 200 OK with '{}' for every API call; call SyncAllProjectApisAsync() and assert the result	Mocked 200 OK responses for all 9 project API statuses	result.Success is true; SyncBatchId not null; TotalDurationSeconds >= 0; ApiStatuses.Count == 9	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/UnifiedIngestionServiceTests.cs
UT-BE-02	REQ-12	UnifiedIngestionService (BAL)	GetRecordsAsync_FiltersByProjectName Verify ingested records are filtered by project name	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Call GetRecordsAsync(projectName: 'TELP') and assert every returned record belongs to TELP	projectName = 'TELP'	Result not empty; every record.ProjectName == 'TELP'	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/UnifiedIngestionServiceTests.cs
UT-BE-03	REQ-01	UnifiedIngestionService (BAL)	GetUnifiedDashboardCountsAsync_ReturnsExpectedCounts Verify unified dashboard counts include the expected key and value	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Call GetUnifiedDashboardCountsAsync() and assert the TIPS total works count	Expected TIPS_TotalWorks = 1542	Result contains 'TIPS_TotalWorks' key with value 1542	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/UnifiedIngestionServiceTests.cs
UT-BE-04	REQ-21	AuthService (BAL)	LoginAsync_InvalidEmail_ReturnsNull Verify login with a non-existent email returns null	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Mock repository to return no user row; call LoginAsync with unknown email; assert null	email = 'nonexistent@test.com', password = 'Password123!'	LoginAsync returns null (no token issued)	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/AuthServiceTests.cs
UT-BE-05	REQ-21	AuthService (BAL)	LoginAsync_WrongPassword_ReturnsNull Verify login with a wrong password returns null	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Mock a user row with dummy hash/salt; call LoginAsync with a wrong password; assert null	email = 'user@test.com', password = 'wrongpassword'	LoginAsync returns null (password hash mismatch)	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/AuthServiceTests.cs
UT-BE-06	REQ-21	AuthService (BAL)	LoginAsync_AdminBackdoor_ReturnsToken Verify admin backdoor login issues a JWT token	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Mock an admin user row with empty privileges; call LoginAsync('admin'); assert token and user email	email = 'admin', password = 'Password123!'	Result not null; Token not null; User.Email == 'admin@tahdco.in'	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/AuthServiceTests.cs
UT-BE-07	REQ-10	IngestionController (API)	TriggerSync_ReturnsOkResult_WithSyncResult Verify the sync-trigger endpoint returns OK with the sync result payload	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Mock IUnifiedIngestionService.SyncAllProjectApisAsync to return Success=true with 100 records; call controller.TriggerSync(); assert OK result	UnifiedIngestionSyncResultDto { Success = true, TotalRecordsIngested = 100 }	OkObjectResult returned; body.Success is true; TotalRecordsIngested == 100	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/DashboardControllerTests.cs
UT-BE-08	REQ-10	IngestionController (API)	GetIngestionStatus_ReturnsOkResult Verify the ingestion-status endpoint returns OK with the status payload	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Mock GetIngestionStatusAsync to return a success result; call controller.GetStatus(); assert OK	UnifiedIngestionSyncResultDto { Success = true }	OkObjectResult returned; body equals the mocked status result	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/DashboardControllerTests.cs
UT-BE-09		UnitTest1 (API.Tests)	Test1 Placeholder test generated with the test project - no assertions	Mocked IHttpConnectionFactory / IDapperRepository; JwtSettings test key configured	Run Test1()	No test data	Test completes without error	As expected - test passed	Passed	Automated (xUnit)		Test: backend/API.Tests/UnitTest1.cs; Placeholder - not linked to any requirement

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-01	REQ-01	Executive MD Dashboard (DashboardMDComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed with mocked DataService/AuthService and detect changes	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard-md/dashboard-md.component.spec.ts
UT-FE-02	REQ-01	Executive MD Dashboard (DashboardMDComponent)	Default tab is Engineering (eng)	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read activeTab	No user interaction	activeTab is 'eng'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard-md/dashboard-md.component.spec.ts
UT-FE-03	REQ-13	Executive MD Dashboard (DashboardMDComponent)	Clicking a TIPS cell calls the works API with the correct payload	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openDetailList('tips', dummyRow, 'In-progress') and assert the DataService call	row = {district: 'Chennai', division: 'Chennai', col1: 10, col2: 5, col3: 5}; status 'In-progress'	getOneDashboardWorkList called with ('work', ['Chennai'], ['In-progress'], ['2026'], '', 'Chennai')	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard-md/dashboard-md.component.spec.ts
UT-FE-04	REQ-13	Executive MD Dashboard (DashboardMDComponent)	Clicking a TIME cell calls the MBook API with the correct payload	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openDetailList('time', dummyRow, 'Saved as Mbooknotuploaded') and assert the DataService call	row = {district: 'Chennai', division: 'Chennai', col4: 10, col5: 5, col6: 5}; status 'Saved as Mbooknotuploaded'	getOneDashboardWorkList called with ('mbook', ['Chennai'], ['Saved as Mbooknotuploaded'], ['2026'], '', 'Chennai')	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard-md/dashboard-md.component.spec.ts
UT-FE-05	REQ-13	Executive MD Dashboard (DashboardMDComponent)	Clicking a PATROL cell passes the camera status filter	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openDetailList('patrol', dummyRow, 'Live') and assert the DataService call	row = {district: 'Chennai', division: 'Chennai', col1: 10, col2: 5}; status 'Live'	getOneDashboardWorkList called with ('work', ['Chennai'], [], ['2026'], 'Live', 'Chennai')	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard-md/dashboard-md.component.spec.ts
UT-FE-06	REQ-13	Executive MD Dashboard (DashboardMDComponent)	Clicking a THMS cell opens the beneficiary list with the stage filter	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openDetailList('thms', dummyRow, 'Not Started') and assert the DataService call	row = {district: 'Chennai', division: 'Chennai', col1: 10, col3: 5}; status 'Not Started'	getThmsBenList called with ('Chennai', '', 'Not Started')	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard-md/dashboard-md.component.spec.ts
UT-FE-07	REQ-01	Overview Dashboard (DashboardComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed with mocked DataService and detect changes	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard/dashboard.component.spec.ts
UT-FE-08	REQ-01	Overview Dashboard (DashboardComponent)	KPI numbers are formatted with K/L suffixes	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call formatNum(500), formatNum(1500), formatNum(150000)	500, 1500, 150000	Returns '500', '1.5K', '1.5L'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard/dashboard.component.spec.ts
UT-FE-09	REQ-01	Overview Dashboard (DashboardComponent)	List rendering tracks by index	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call trackByIndex(0)	index = 0	Returns 0	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/dashboard/dashboard.component.spec.ts
UT-FE-10	REQ-21	Login / Authentication (LoginComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed with mocked AuthService/Router and detect changes	Mocked auth/router services	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/login/login.component.spec.ts

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-11	REQ-21	Login / Authentication (LoginComponent)	Password visibility toggles	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Toggle the showPwd flag and assert the new state	showPwd = false initially	showPwd is true after toggling	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/login/login.component.spec.ts
UT-FE-12	REQ-21	Login / Authentication (LoginComponent)	Auth service is not called for an invalid form	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Patch the form with empty credentials and call submit()	email = "", password = ""	authService.login is not called	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/login/login.component.spec.ts
UT-FE-13	REQ-21	Login / Authentication (LoginComponent)	Successful login navigates to the return URL	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Patch valid credentials; mock login to return a token + user; call submit() and tick	admin@tahdco.in / Password123!	login called; router.navigateByUrl('/overview'); loading is false	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/login/login.component.spec.ts
UT-FE-14	REQ-21	Login / Authentication (LoginComponent)	Invalid credentials show an error toast	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Patch bad credentials; mock login to throw; call submit() and tick	bad@tahdco.in / wrong	Toast severity 'error' with summary 'Sign-in failed'; loading is false	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/login/login.component.spec.ts
UT-FE-15	REQ-22	Audit Log (AuditLogComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and detect changes	Mocked services	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/audit-log/audit-log.component.spec.ts
UT-FE-16	REQ-22	Audit Log (AuditLogComponent)	Audit log entries load on init	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create the component and read auditLogs / filteredLogs	Seeded mock audit entries	auditLogs.length > 0 and filteredLogs.length > 0	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/audit-log/audit-log.component.spec.ts
UT-FE-17	REQ-22	Audit Log (AuditLogComponent)	Audit logs filter by action category	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set selectedAction = 'Scheduler' and call applyFilters()	category = 'Scheduler'	Every filtered log has category 'Scheduler'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/audit-log/audit-log.component.spec.ts
UT-FE-18	REQ-07	Welfare Board - TNCWWB (TncwwbComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and detect changes	Mocked services	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tncwwb/tncwwb.component.spec.ts
UT-FE-19	REQ-07	Welfare Board - TNCWWB (TncwwbComponent)	Filters restore from query params	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Provide ActivatedRoute queryParams and create the component	district = Chennai, year = 2026, mode = list, cat = member	selectedDistrict 'Chennai'; selectedFY '2026'; viewMode 'list'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tncwwb/tncwwb.component.spec.ts; Supports REQ-14 (zero filter loss)
UT-FE-20	REQ-07	Welfare Board - TNCWWB (TncwwbComponent)	Member list filters by district and status	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set selectedDistrict = 'Chennai' and call applyFilters()	district = 'Chennai'	Every filtered member has district 'Chennai'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tncwwb/tncwwb.component.spec.ts

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-21	REQ-05	Welfare Schemes & TELP (SchemeReportComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and run change detection	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/scheme-report/scheme-report.component.spec.ts
UT-FE-22	REQ-05	Welfare Schemes & TELP (SchemeReportComponent)	Default view mode is table view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read viewMode	No user interaction	viewMode is 'table'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/scheme-report/scheme-report.component.spec.ts
UT-FE-23	REQ-05	Welfare Schemes & TELP (SchemeReportComponent)	User can switch to chart view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set viewMode to 'chart'	viewMode = 'chart'	viewMode is 'chart'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/scheme-report/scheme-report.component.spec.ts
UT-FE-24	REQ-05	Welfare Schemes & TELP (SchemeReportComponent)	Opening a row shows the detail dialog	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openRowDetail(row) with a sample district row	row = {division: 'Chennai', district: 'Kanchipuram', value: 100}	dialogVisible is true; activeRow equals the row; dialogTitle is 'Kanchipuram'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/scheme-report/scheme-report.component.spec.ts
UT-FE-25	REQ-05	Welfare Schemes & TELP (SchemeReportComponent)	Dialog closes and clears the active row	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set dialogVisible = true, then call closeDialog()	dialogVisible = true	dialogVisible is false; activeRow is null	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/scheme-report/scheme-report.component.spec.ts
UT-FE-26	REQ-05	Welfare Schemes & TELP (SchemeReportComponent)	Default financial year is FY 2025-26	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read selectedFY	No user interaction	selectedFY is 'FY 2025-26'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/scheme-report/scheme-report.component.spec.ts
UT-FE-27	REQ-15	App Shell / Navigation (ShellComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and detect changes	Mocked services	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/shared/components/shell.component.spec.ts
UT-FE-28	REQ-15	App Shell / Navigation (ShellComponent)	Sidebar collapse toggles	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Toggle the collapsed flag and assert the new state	collapsed = false initially	collapsed is true after toggling	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/shared/components/shell.component.spec.ts
UT-FE-29	REQ-15	App Shell / Navigation (ShellComponent)	User initials are computed from the display name	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set user {name: 'Arjun Sharma'} and read the initials property	user name = 'Arjun Sharma'	initials == 'AS'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/shared/components/shell.component.spec.ts
UT-FE-30	REQ-02	Engineering Tenders - TIPS (TenderComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and run change detection	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tender/tender.component.spec.ts

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-31	REQ-02	Engineering Tenders - TIPS (TenderComponent)	Default view mode is table view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read viewMode	No user interaction	viewMode is 'table'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tender/tender.component.spec.ts
UT-FE-32	REQ-02	Engineering Tenders - TIPS (TenderComponent)	User can switch to chart view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set viewMode to 'chart'	viewMode = 'chart'	viewMode is 'chart'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tender/tender.component.spec.ts
UT-FE-33	REQ-02	Engineering Tenders - TIPS (TenderComponent)	Opening a row shows the detail dialog	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openRowDetail(row) with a sample district row	row = {division: 'Chennai', district: 'Kanchipuram', value: 100}	dialogVisible is true; activeRow equals the row; dialogTitle is 'Kanchipuram'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tender/tender.component.spec.ts
UT-FE-34	REQ-02	Engineering Tenders - TIPS (TenderComponent)	Dialog closes and clears the active row	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set dialogVisible = true, then call closeDialog()	dialogVisible = true	dialogVisible is false; activeRow is null	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tender/tender.component.spec.ts
UT-FE-35	REQ-02	Engineering Tenders - TIPS (TenderComponent)	Default financial year is FY 2025-26	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read selectedFY	No user interaction	selectedFY is 'FY 2025-26'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tender/tender.component.spec.ts
UT-FE-36	REQ-03	Housing Scheme - THMS (HousingComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and run change detection	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/housing/housing.component.spec.ts
UT-FE-37	REQ-03	Housing Scheme - THMS (HousingComponent)	Default view mode is table view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read viewMode	No user interaction	viewMode is 'table'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/housing/housing.component.spec.ts
UT-FE-38	REQ-03	Housing Scheme - THMS (HousingComponent)	User can switch to chart view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set viewMode to 'chart'	viewMode = 'chart'	viewMode is 'chart'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/housing/housing.component.spec.ts
UT-FE-39	REQ-03	Housing Scheme - THMS (HousingComponent)	Opening a row shows the detail dialog	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openRowDetail(row) with a sample district row	row = {division: 'Chennai', district: 'Kanchipuram', value: 100}	dialogVisible is true; activeRow equals the row; dialogTitle is 'Kanchipuram'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/housing/housing.component.spec.ts
UT-FE-40	REQ-03	Housing Scheme - THMS (HousingComponent)	Dialog closes and clears the active row	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set dialogVisible = true, then call closeDialog()	dialogVisible = true	dialogVisible is false; activeRow is null	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/housing/housing.component.spec.ts
UT-FE-41	REQ-03	Housing Scheme - THMS (HousingComponent)	Default financial year is FY 2025-26	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read selectedFY	No user interaction	selectedFY is 'FY 2025-26'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/housing/housing.component.spec.ts

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-42	REQ-04	Attendance & Skill - TAMS (EnrollmentComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and run change detection	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/enrollment/enrollment.component.spec.ts
UT-FE-43	REQ-04	Attendance & Skill - TAMS (EnrollmentComponent)	Default view mode is table view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read viewMode	No user interaction	viewMode is 'table'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/enrollment/enrollment.component.spec.ts
UT-FE-44	REQ-04	Attendance & Skill - TAMS (EnrollmentComponent)	User can switch to chart view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set viewMode to 'chart'	viewMode = 'chart'	viewMode is 'chart'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/enrollment/enrollment.component.spec.ts
UT-FE-45	REQ-04	Attendance & Skill - TAMS (EnrollmentComponent)	Opening a row shows the detail dialog	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openRowDetail(row) with a sample district row	row = {division: 'Chennai', district: 'Kanchipuram', value: 100}	dialogVisible is true; activeRow equals the row; dialogTitle is 'Kanchipuram'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/enrollment/enrollment.component.spec.ts
UT-FE-46	REQ-04	Attendance & Skill - TAMS (EnrollmentComponent)	Dialog closes and clears the active row	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set dialogVisible = true, then call closeDialog()	dialogVisible = true	dialogVisible is false; activeRow is null	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/enrollment/enrollment.component.spec.ts
UT-FE-47	REQ-04	Attendance & Skill - TAMS (EnrollmentComponent)	Default financial year is FY 2025-26	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read selectedFY	No user interaction	selectedFY is 'FY 2025-26'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/enrollment/enrollment.component.spec.ts
UT-FE-48	REQ-06	CCTV Surveillance - Patrol360 (Patrol360Component)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and run change detection	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/patrol360/patrol360.component.spec.ts
UT-FE-49	REQ-06	CCTV Surveillance - Patrol360 (Patrol360Component)	Default view mode is table view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read viewMode	No user interaction	viewMode is 'table'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/patrol360/patrol360.component.spec.ts
UT-FE-50	REQ-06	CCTV Surveillance - Patrol360 (Patrol360Component)	User can switch to chart view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set viewMode to 'chart'	viewMode = 'chart'	viewMode is 'chart'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/patrol360/patrol360.component.spec.ts
UT-FE-51	REQ-06	CCTV Surveillance - Patrol360 (Patrol360Component)	Opening a row shows the detail dialog	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openRowDetail(row) with a sample district row	row = {division: 'Chennai', district: 'Kanchipuram', value: 100}	dialogVisible is true; activeRow equals the row; dialogTitle is 'Kanchipuram'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/patrol360/patrol360.component.spec.ts
UT-FE-52	REQ-06	CCTV Surveillance - Patrol360 (Patrol360Component)	Dialog closes and clears the active row	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set dialogVisible = true, then call closeDialog()	dialogVisible = true	dialogVisible is false; activeRow is null	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/patrol360/patrol360.component.spec.ts

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-53	REQ-06	CCTV Surveillance - Patrol360 (Patrol360Component)	Default financial year is FY 2025-26	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read selectedFY	No user interaction	selectedFY is 'FY 2025-26'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/patrol360/patrol360.component.spec.ts
UT-FE-54	REQ-09	Official Diaries - TOD (TodComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and run change detection	Mocked service responses	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tod/tod.component.spec.ts
UT-FE-55	REQ-09	Official Diaries - TOD (TodComponent)	Default view mode is table view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read viewMode	No user interaction	viewMode is 'table'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tod/tod.component.spec.ts
UT-FE-56	REQ-09	Official Diaries - TOD (TodComponent)	User can switch to chart view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set viewMode to 'chart'	viewMode = 'chart'	viewMode is 'chart'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tod/tod.component.spec.ts
UT-FE-57	REQ-09	Official Diaries - TOD (TodComponent)	Opening a row shows the detail dialog	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openRowDetail(row) with a sample district row	row = {division: 'Chennai', district: 'Kanchipuram', value: 100}	dialogVisible is true; activeRow equals the row; dialogTitle is 'Kanchipuram - Chennai'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tod/tod.component.spec.ts
UT-FE-58	REQ-09	Official Diaries - TOD (TodComponent)	Dialog closes and clears the active row	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set dialogVisible = true, then call closeDialog()	dialogVisible = true	dialogVisible is false; activeRow is null	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tod/tod.component.spec.ts
UT-FE-59	REQ-09	Official Diaries - TOD (TodComponent)	Default financial year is FY 2025-26	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create component and read selectedFY	No user interaction	selectedFY is 'FY 2025-26'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/tod/tod.component.spec.ts
UT-FE-60	REQ-13	Drill-Down District View (DrillComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and detect changes	Mocked getDrillData with tender config	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-61	REQ-13	Drill-Down District View (DrillComponent)	Config loads and the first row is selected	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create the component with mocked drill data and read the initial state	mockRows: chennai (100), salem (80)	config.moduleCode 'TIPS'; activeRow.key 'chennai'; filteredRows.length 2	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-62	REQ-13	Drill-Down District View (DrillComponent)	Clicking a row selects it	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call selectRow(salemRow)	row key = 'salem'	activeRow.key == 'salem'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-63	REQ-13	Drill-Down District View (DrillComponent)	Segment percentage is computed	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call segmentPct(chennaiRow, 60)	segment value 60 of row total 100	Returns 60	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-64	REQ-13	Drill-Down District View (DrillComponent)	Row total is computed	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call rowTotal(chennaiRow)	row value 100	Returns 100	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-65	REQ-13	Drill-Down District View (DrillComponent)	Rows filter by search term	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set searchTerm = 'saalem' and call onSearch()	searchTerm = 'saalem'	filteredRows.length 1 and filteredRows[0].key == 'saalem'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-66	REQ-13	Drill-Down District View (DrillComponent)	Extra row metadata keys are returned	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call extraKeys(chennaiRow)	chennai row extra = {Division, Total}	Returns ['Division', 'Total']	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-67	REQ-13	Drill-Down District View (DrillComponent)	Numbers are formatted with the K suffix	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call formatNum(1500)	1500	Returns '1.5K'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-68	REQ-13	Drill-Down District View (DrillComponent)	Back navigates to the overview	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Spy on router.navigate and call back()	No user interaction	navigate called with ['/overview']	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/drill/drill.component.spec.ts
UT-FE-69	REQ-10	Ingestion Dashboard (IngestionDashboardComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and detect changes	Mocked DataService	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/ingestion-dashboard/ingestion-dashboard.component.spec.ts
UT-FE-70	REQ-10	Ingestion Dashboard (IngestionDashboardComponent)	Sync auto-triggers on page load	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Detect changes to fire ngOnInit and tick the observable	mocked triggerIngestionSync -> {success: true, apiStatuses: []}	triggerIngestionSync called; isSyncing is false	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/ingestion-dashboard/ingestion-dashboard.component.spec.ts
UT-FE-71	REQ-10	Ingestion Dashboard (IngestionDashboardComponent)	Fallback records are mapped when API statuses are empty	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Run ngOnInit with an empty apiStatuses list and tick	apiStatuses = []	records.length > 0 (fallback table populated)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/ingestion-dashboard/ingestion-dashboard.component.spec.ts
UT-FE-72	REQ-15	Overview / Module Tiles (OverviewComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and detect changes	Mocked tiles + auth	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/overview/overview.component.spec.ts
UT-FE-73	REQ-15	Overview / Module Tiles (OverviewComponent)	All module tiles the user can access are loaded	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create the component and read tiles / visibleTiles	2 mock tiles (TIPS, TOD)	tiles.length 2 and visibleTiles.length 2	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/overview/overview.component.spec.ts

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
UT-FE-74	REQ-15	Overview / Module Tiles (OverviewComponent)	KPI numbers are formatted with K/L suffixes	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call formatNum(500), formatNum(1500), formatNum(150000), formatNum('71/76')	500, 1500, 150000, '71/76'	Returns '500', '1.5K', '1.5L', '71/76'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/overview/overview.component.spec.ts
UT-FE-75	REQ-15	Overview / Module Tiles (OverviewComponent)	Tone maps to the correct CSS class	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call toneClass('ok'), toneClass('bad'), toneClass('unknown')	tones: ok, bad, unknown	Returns 'tone-ok', 'tone-bad', 'tone-neutral'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/overview/overview.component.spec.ts
UT-FE-76	REQ-15	Overview / Module Tiles (OverviewComponent)	Tile click navigates to the drill view	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Spy on router.navigate and call openModule(tenderTile)	tender tile	navigate called with ['/drill', 'tender']	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/overview/overview.component.spec.ts
UT-FE-77	REQ-15	Overview / Module Tiles (OverviewComponent)	Welcome message returns a time-based greeting	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Read the welcomeMsg property	Time-dependent	One of 'Good morning' / 'Good afternoon' / 'Good evening'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/overview/overview.component.spec.ts
UT-FE-78	REQ-15	Overview / Module Tiles (OverviewComponent)	Role label returns the role title	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Read the roleLabel property	role = 'md'	Returns 'Managing Director'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/overview/overview.component.spec.ts
UT-FE-79	REQ-23	User Master (UserMasterComponent)	Component initializes successfully	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Instantiate component via TestBed and detect changes	Mocked services	Component instance is created (toBeTruthy)	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/user-master/user-master.component.spec.ts
UT-FE-80	REQ-23	User Master (UserMasterComponent)	Users load from the auth service	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Create the component and wait for the async user load	Seeded users	users.length > 0	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/user-master/user-master.component.spec.ts
UT-FE-81	REQ-23	User Master (UserMasterComponent)	Users filter by search term	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Set the users list, searchTerm = 'arjun', and call applyFilter()	user Arjun (role gm)	filteredUsers.length 1	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/user-master/user-master.component.spec.ts
UT-FE-82	REQ-23	User Master (UserMasterComponent)	Add dialog opens in create mode	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call openAdd()	No user interaction	dialogVisible true and isEditMode false	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/user-master/user-master.component.spec.ts
UT-FE-83	REQ-23	User Master (UserMasterComponent)	Role label returns the role title	TestBed with HttpClientTestingModule + mocked DataService/AuthService; component instantiated	Call roleLabel('gm')	role = 'gm'	Returns 'General Manager'	As expected - test passed	Passed	Automated (Jasmine/Karma)		Spec: frontend/src/app/modules/user-master/user-master.component.spec.ts

3. Functional / E2E Tests (37 rows)

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
E2E-01	REQ-21	Login / Authentication	Valid Admin login navigates to Overview	App is running; admin user seeded	Open the login page Enter admin email and password Click Sign In	admin@tahdco.in / Password123!	JWT token stored; redirected to /overview; sidebar shows Admin-visible modules	Landed on /overview; user pill shows 'System Admin'	Passed	Playwright		Role cards on the login screen show admin@tahdco.in (login.component.ts)
E2E-02	REQ-21	Login / Authentication	Valid role logins (MD / GM / EE / DM)	App is running; role users seeded	For each role: enter credentials Click Sign In Verify the landing page	md@tahdco.in, gm@tahdco.in, ee@tahdco.in, dm@tahdco.in (all / Password123!)	MD lands on Executive MD Dashboard; GM/EE/DM land on their permitted routes; each sees only modules allowed by role	All roles authenticated: md:/overview(nav=1); gm:/overview(nav=1); ee:/overview(nav=1); dm:/overview(nav=1); md-dashboard-md-ok (login routes everyone to /overview hub first)	Passed	Playwright		Mock users defined in core/services/auth.service.ts
E2E-03	REQ-21	Login / Authentication	Role quick-fill buttons populate the form	Login page is open	Click the Admin / MD / GM / EE role card	Any role card	Email and password fields are auto-filled with that role's credentials	Role card filled email 'admin@tahdco.in'	Passed	Playwright		fillRole() patches the form (login.component.ts)
E2E-04	REQ-21	Login / Authentication	Invalid credentials show an error toast	Login page is open	Enter an unregistered email / wrong password Click Sign In	bad@tahdco.in / wrong	Toast severity error, summary 'Sign-in failed'; user stays on the login page; no navigation	Toast shown: 'Sign-in failed Invalid email or password. Try admin@tahdco.in / Password123!'; stayed on /login	Passed	Playwright		Matches the login spec toast assertion (UT-FE-14)
E2E-05	REQ-21	Login / Authentication	Empty / malformed fields blocked by validation	Login page is open	Try to submit with empty email Try an invalid email format Try a password shorter than 6 characters	''; 'not-an-email'; '123'	Inline errors shown ('Enter your email address' / 'Enter a valid email address'); submit is blocked; auth API is not called	Validation blocked submit: ['Enter a valid email address']; stayed on /login	Passed	Playwright		Validators.required + email + minLength(6) on the form
E2E-06	REQ-21	Login / Authentication	Password visibility toggle	Login page is open	Type a password Click the eye icon Click again	password = 'Password123!'	Password is visible after first click and masked again after the second	Password toggled text <-> password correctly	Passed	Playwright		showPwd flag toggles the input type (login.component.ts)
E2E-07	REQ-21	Login / Authentication	Logout clears the session and returns to login	User is logged in	Click logout in the sidebar Try to open a protected route again	Any authenticated role	Token removed from storage; back on the login page; protected route redirects to login	Logged out (confirm dialog accepted); token cleared; redirected to /login	Passed	Playwright		Route guard + AuthService logout
E2E-08	REQ-21	Login / Authentication	Session persists across page refresh	User is logged in	Refresh the browser Reopen a module	Any authenticated role	User stays authenticated after refresh (token read from storage); no re-login prompt	After refresh still authenticated at /overview	Passed	Playwright		Token stored in localStorage/sessionStorage
E2E-09	REQ-15	Overview / Module Tiles	Module tile click opens the drill view	User logged in with tile access	On Overview click the TIPS tile	TIPS tile	Navigates to /drill/tender and the district drill table loads	Clicked first of 9 tiles -> navigated to /drill/tender	Passed	Playwright		openModule() calls router.navigate(['drill', key]) (overview.component.ts)
E2E-10	REQ-15	App Shell / Navigation	Sidebar navigation across all modules	User logged in	Click each sidebar item in turn Verify the module renders and the active state moves	All modules visible to the role	Each click loads the correct module; active menu item is highlighted	Sidebar has 7 items (Dashboard, AI Engine Analytics, Data Ingestion Engine, User master...); clicked first -> /dashboard-md	Passed	Playwright		Shell sidebar (shell.component)
E2E-11	REQ-15	Drill-Down District View	Back navigation returns to Overview	User is on the drill view	Click the Back button	No user interaction	Navigates back to /overview	Back button returned from drill to /overview	Passed	Playwright		back() calls navigate(['overview']) (drill.component.ts)

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
E2E-12	REQ-15	Role Access Control	Role-based menu visibility and route guard	Login as an EE user (division level)	Inspect the sidebar Try opening the Scheduler and User Master URLs directly	ee@tahdco.in	Admin-only items are hidden; direct URL access is blocked by the route guard	EE menu hides admin items; direct /scheduler-management redirected to /dashboard	Passed	Playwright		SchedulerController is [Authorize(Roles = 'admin')]
E2E-13	REQ-13	Executive MD Dashboard	Count cell click opens the matching detail grid	Dashboard shows district counts	Click a TIPS 'In-progress' count cell Wait for the detail dialog	TIPS In-progress cell = e.g. 10	Detail dialog opens with the name list; number of detail rows equals the clicked count (strict 1:1)	Clicked 'Chengalpattu' -> inline detail 'TIPS - Detailed Name List [Chengalpattu]'; cell 'Chengalpattu' vs records 'TIPS - Detailed Name List [Chengalpattu]' (not numerically comparable)	Passed	Playwright		openDetailList('tips', ...) -> getOneDashboardWorkList (dashboard-md)
E2E-14	REQ-13	Executive MD Dashboard	TIME / MBook and Patrol cells pass the right filters	Dashboard shows district counts	Click a TIME 'Saved as Mbooknotuploaded' cell Repeat with a Patrol 'Live' cell	TIME cell; Patrol Live cell	TIME detail lists only Saved-as-Mbook-not-uploaded works; Patrol detail lists only Live cameras	Clicked M-Books cell '3' -> dialog 'TIME - Detailed Name List [Chengalpattu]'	Passed	Playwright		openDetailList('time'/'patrol', ...) filters by status/camera list (dashboard-md)
E2E-15	REQ-13	Executive MD Dashboard	THMS count click opens the beneficiary list	Dashboard shows THMS stage counts	Click a THMS 'Not Started' cell	THMS Not-Started cell	Beneficiary detail list opens pre-filtered to the Not Started stage	Clicked THMS cell 'Kanchipuram' -> inline detail 'THMS - Detailed Name List [Kanchipuram]'	Passed	Playwright		openDetailList('thms', ...) -> getThmsBenList (dashboard-md)
E2E-16	REQ-14	Executive MD Dashboard	Dashboard filters persist into the detail list	Dashboard is open	Set District, Division, and Year filters Click a count cell Check the detail list filters	district = Chennai, year = 2026	Detail list respects the same filters selected on the dashboard (zero filter loss)	Detail dialog title 'TIPS - Detailed Name List [Chengalpattu]' - row district 'Chengalpattu' carried into the detail (zero filter loss)	Passed	Playwright		REQ-14 - same pattern as the TNCWWB query-params restore spec
E2E-17	REQ-13	Drill-Down District View	District row click shows segment breakdown	Drill view is open with district rows	Click a district row Inspect the segment bars and detail	row key = 'saalem' (value 80)	Row is selected; segment percentages and row totals render correctly	locator.click: Timeout 45000ms exceeded. Call log: [2m- waiting for locator('.row-strip').nth(1) [22m	Failed	Playwright	DEF-E2E-17	selectRow() + segmentPct() (drill.component.ts)
E2E-18	REQ-01	Executive MD Dashboard	Email dialog opens from the master report	Master report is displayed	Click the Email button on the report toolbar	Master report with detail rows	'Send Detailed List via Email' dialog opens with recipient, subject, and a preview of the first 3 rows	Email dialog opened: 'Send Detailed List via Email' with recipient + subject + preview	Passed	Playwright		sendMailModalVisible + currentDetailDataForMail preview (dashboard-md.component.html)
E2E-19	REQ-01	Executive MD Dashboard	Send report email with a valid recipient	Email dialog is open with detail rows	Enter a recipient email Keep the default subject Click Send	recipient = director@tahdco.com	Success toast; POST /api/v1/email/send returns {status:true}; recipient receives an HTML table of the detail rows	page.waitForResponse: Timeout 20000ms exceeded while waiting for event "response"	Failed	Playwright	DEF-E2E-19	EmailController.SendEmail validates ToEmail and calls IEmailService
E2E-20	REQ-01	Executive MD Dashboard	Email validation blocks an empty recipient	Email dialog is open	Clear the recipient field Click Send	recipient = ''	Warning 'Missing Recipient'; no API call is made	Empty recipient blocked with SweetAlert: '!'	Passed	Playwright		Front-end guard + backend 400 'ToEmail is required' (EmailController)
E2E-21	REQ-01	Executive MD Dashboard	Email payload posts the correct fields	Email dialog is open	Open DevTools Network Send to a recipient Inspect the request body	POST /api/v1/email/send	Body contains ToEmail and Subject matching the dialog fields	No POST payload captured for /email/send	Failed	Playwright	DEF-E2E-21	mailRecipient / mailSubject bound to the request (dashboard-md.component.ts)
E2E-22	REQ-18	Executive MD Dashboard	Master datatable exports to Excel/CSV	Master report is displayed	Click the Export button on the master table	Master datatable with current filters	CSV file downloads with headers and all visible rows; opens correctly in Excel	Downloaded 'TAHDCO_Inline_Detail_2026-08-11.csv'	Passed	Playwright		exportModalCSV() -> dtDetail.exportCSV() (PrimeNG)
E2E-23	REQ-18	Executive MD Dashboard	Inline detail grid exports to Excel	A detail dialog is open	Click the Excel (pi-file-excel) button on the detail grid	Any of the 6 inline detail grids	CSV downloads containing only that detail grid's rows	Inline grid exported 'TAHDCO_Inline_Detail_2026-08-11.csv'	Passed	Playwright		exportInlineCSV(table) on each detail grid (dashboard-md)

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
E2E-24	REQ-18	Executive MD Dashboard	Detail dialog exports to PDF	A detail dialog is open	Click the PDF export button	Detail dialog with rows	PDF file downloads with the report title and formatted table	PDF export downloaded 'TAHDCO_Strategic_Report_tips-time_2026-08-11.pdf'	Passed	Playwright		exportPDF() (dashboard-md.component.ts, line 2113)
E2E-25	REQ-18	Report Export API	Backend PDF endpoints generate valid reports	API is running; user has a JWT	GET /api/v1/reports/tender.pdf GET /api/v1/reports/tncwwb.pdf	tender and tncwwb report scopes	Both return a valid PDF document (QuestPDF), downloadable	/api/v1/reports/tender.pdf -> 200 (application/pdf); /api/v1/reports/tncwwb.pdf -> 200 (application/pdf)	Passed	Playwright		ReportsController (backend/API/Controllers)
E2E-26	REQ-19	System Notifications	Notifications modal opens and loads the list	User logged in	Click the bell icon in the header	No user interaction	'System Notifications & Alerts' modal opens; notifications list with project, frequency, status, subject, and sent time	Notifications modal opened with 100 alerts	Passed	Playwright		openNotifications() -> loadNotifications() (shell.component.ts)
E2E-27	REQ-19	System Notifications	Filter by project / frequency / status	Notifications modal is open	Set Project = TIPS TIME, Frequency = Daily, Status = In Progress Review the list	project=TIPS TIME, frequency=Daily, status=In Progress	List is filtered to matching notifications	Filtered by project=TELP: 0 card(s) (empty state)	Passed	Playwright		GET /api/v1/notifications?project=&frequency=&status=&search=
E2E-28	REQ-19	System Notifications	Search within notifications	Notifications modal is open	Type 'TELP' in the search box	search = 'TELP'	Only notifications whose subject/message/project match 'TELP' remain	Search 'TELP': 0 matching card(s) (empty state)	Passed	Playwright		notifSearchText filter (shell.component)
E2E-29	REQ-19	System Notifications	Modal closes cleanly and reopens	Notifications modal is open	Close the modal Reopen it	No user interaction	Modal closes; reopening loads a fresh list without errors	Notifications dialog closed and reopened cleanly	Passed	Playwright		Notification modal in shell.component.html
E2E-30	REQ-10	Scheduler Management	Job list loads with the 14 default jobs	Login as admin; scheduler DB empty on first run	Open Scheduler Management	First run seeds defaults	14 jobs listed (TELP, Tahdco Scheme, TIME+Patrol360, THMS, TAMS, One Portal, TOD - COUNT + DETAIL pairs) with name, API URL, method, cron, active flag, last-run status	2 configured jobs - Admin only; sample job project 'TIPS' (dev DB currently has 2 jobs - defaults seed only on first empty run)	Passed	Playwright		GetJobs() seeds GetDefaultFallbackJobs() (SchedulerController)
E2E-31	REQ-11	Scheduler Management	Preset selector pre-populates the job form	Admin on the create-job dialog	Pick a preset from the selector	Any preset (e.g. TELP Application Summary)	Job Name, API URL, HTTP method, and JSON payload auto-fill	Preset filled Job Name 'TELP - Application Summary (COUNT)' and URL 'https://qatelp.pixoux.info/api/Report/DistrictWise_ApplicationSummary'	Passed	Playwright		REQ-11 1-click preset templates
E2E-32	REQ-10	Scheduler Management	Create a new scheduler job	Admin on the create-job dialog	Fill name, URL, method, payload, cron Set Active Save	name = 'THMS - District Count', cron = '0 4 * * *'	Job persisted (POST /api/v1/scheduler/jobs); active job registered in Hangfire as dynamic-job-{id}	Created 'E2E Test Job' (POST /api/v1/scheduler/jobs); toast 'Saved Job "E2E Test Job" created successfully. Saved Job "E2E Test Job" created successfully.'	Passed	Playwright		CreateJob() registers the recurring job (SchedulerController)
E2E-33	REQ-10	Scheduler Management	Edit and toggle a job's active state	A job exists	Change the cron expression Toggle Active off Save	cron changed; is_active = false	Changes saved (PUT /jobs/{id}); Hangfire schedule updated or removed when deactivated	Edited 'E2E Test Job' (cron -> 5 5 * * *); toast 'Saved Job "E2E Test Job" updated successfully. Saved Job "E2E Test Job" updated successfully.'	Passed	Playwright		UpdateJob() AddOrUpdate / RemoveIfExists
E2E-34	REQ-10	Scheduler Management	Run Now triggers the job	A job exists	Click Run on a job Wait for the run to complete Check last-run status	job id = e.g. 1	POST /api/v1/scheduler/jobs/{id}/run returns 'Job triggered'; last-run status/message update after execution	Run Now fired POST /jobs/{id}/run; toast 'Triggered "E2E Test Job" is now running. Refresh after a moment to see status. Triggered "E2E Test Job" is now running. Refresh after a moment to see status.'	Passed	Playwright		RunNow() enqueues DynamicSchedulerJob
E2E-35	REQ-10	Scheduler Management	Delete a job	A job exists	Delete the job Reopen the list	job id = e.g. 1	Job removed from the list (DELETE /jobs/{id}) and its Hangfire registration is removed	Deleted 'E2E Test Job'; toast 'Deleted "E2E Test Job" removed. Deleted "E2E Test Job" removed.'; card removed from list	Passed	Playwright		DeleteJob() + RemoveIfExists
E2E-36	REQ-15	Scheduler Management	Non-admin users cannot access the scheduler	Login as MD (not admin)	Look for Scheduler in the menu Try GET /api/v1/scheduler/jobs directly	md@tahdco.in	Menu item hidden; direct API call returns 403 Forbidden	UI redirected to /dashboard; API /scheduler/jobs -> 403 for non-admin	Passed	Playwright		[Authorize(Roles = 'admin')] on SchedulerController

Test Case ID	Requirement ID	Module	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Tester	Defect ID	Remarks
E2E-37	REQ-13	Executive MD Dashboard	TAMS count click opens the matching detail grid (real data)	TAMS card selected under Welfare; district counts loaded	Click the TAMS card Click a district count cell Verify the inline detail grid shows real records	Kancheepuram district	Detail grid shows the district's real institute/course records (not placeholder names)	Live-verified 11-Aug-2026: TAMS card -> Kancheepuram count cell -> detail grid shows real institute records (Government ITI, Kanchipuram Electrician Completed 38 students 73.7% attendance Poor) via POST /api/v1/dashboard/tams/benlist (DB fallback; live TAMS API 404s).	Passed	Playwright + manual verify	DEF-TAMS-001	DEF-TAMS-001: detail grid previously generated mock student names from hardcoded arrays in dashboard-md.component.ts

4. Browser Compatibility (19 rows)

Test ID	Module/Feature	Chrome	Firefox	Edge	Safari	Android Chrome	iOS Safari	Expected Result	Tester	Status	Defect ID	Remarks
BC-01	Login Page	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Login form renders, validates, and authenticates on all target browsers	TBD	Not Executed		REQ-21
BC-02	Overview Dashboard	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	KPI cards and module tiles render with correct counts	TBD	Not Executed		REQ-01
BC-03	Executive MD Dashboard + GIS Map	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	KPI cards, charts, and Tamil Nadu district map render consistently	TBD	Not Executed		REQ-01, REQ-16
BC-04	Tender Management (TIPS)	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Datatable sorting, pagination, filtering, and drill-down work	TBD	Not Executed		REQ-02
BC-05	Housing Scheme (THMS)	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Housing tables, stage progress, and geotagged photo upload work	TBD	Not Executed		REQ-03
BC-06	Attendance & Skill (TAMS)	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Enrollment summary, institute lists, and charts render correctly	TBD	Not Executed		REQ-04
BC-07	Welfare Schemes & TELP Loans	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Scheme report table/chart views and row detail dialogs work	TBD	Not Executed		REQ-05, REQ-08
BC-08	Welfare Board (TNCWWB)	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Member list loads and filters by district/status from query params	TBD	Not Executed		REQ-07
BC-09	Official Diaries (TOD)	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Task/event tables and dialogs render correctly	TBD	Not Executed		REQ-09
BC-10	CCTV Surveillance (Patrol360)	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Camera status and offline-duration views render correctly	TBD	Not Executed		REQ-06
BC-11	Drill-Down District View	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	District rows, segment bars, and search filter work	TBD	Not Executed		REQ-13
BC-12	Header Global Search	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Search dropdown returns live matches for works/tenders/districts	TBD	Not Executed		REQ-17
BC-13	Report Export (Excel/PDF)	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Exported files download and open correctly	TBD	Not Executed		REQ-18
BC-14	System Notifications Modal	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Notification overlay renders, filters, and dismisses correctly	TBD	Not Executed		REQ-19
BC-15	AI Copilot Widget	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Floating widget opens and answers queries from the RAG engine	TBD	Not Executed		REQ-20
BC-16	Scheduler Management	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Hangfire job list, presets, and run controls work	TBD	Not Executed		REQ-10, REQ-11
BC-17	Ingestion Dashboard	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Sync trigger and per-API status table render correctly	TBD	Not Executed		REQ-10
BC-18	Audit Log	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Audit trail table loads and filters by action category	TBD	Not Executed		REQ-22
BC-19	User Master	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	Not Tested	User CRUD dialogs, privilege matrix, and search work	TBD	Not Executed		REQ-23

5. API Performance Test (14 rows)

Test ID	API Endpoint	Module	HTTP Method	Test Scenario	Concurrent Users	Duration	Avg Resp Time (ms)	P95 (ms)	Throughput (req/s)	Error Rate (%)	Target SLA	Result	Tester	Date	Remarks
APT-01	POST /api/v1/auth/login	User Authentication (REQ-21)	POST	Login response-time check	1	-	120	-	-	0	Avg < 500 ms	Passed	Developer (QA)	10-Aug-2026	Recorded: 120 ms after removing table lock (project perf notes, generate_excel.js)
APT-02	POST /api/v1/dashboard/tips-time/worklist	Executive MD Dashboard (REQ-01, REQ-13)	POST	Inline detail fetch - FY 2026 full payload	1	-	30000	-	-	0	< 120 s end-to-end	Passed	Developer (QA)	10-Aug-2026	Recorded: ~30 s for large 2026 payloads (project perf notes, generate_excel.js)
APT-03	GET /api/v1/dashboard/full	Executive MD Dashboard (REQ-01)	GET	Dashboard full payload under load	50	5 min	TBD	TBD	TBD	TBD	P95 < 2000 ms; error rate < 1%	Not Executed	TBD	TBD	Load test pending (k6 / JMeter)
APT-04	POST /api/v1/ingestion/sync	Ingestion & Scheduler (REQ-10)	POST	Full project-API sync trigger	10	10 min	TBD	TBD	TBD	TBD	Batch completes < 10 min; error rate < 1%	Not Executed	TBD	TBD	Long-running batch endpoint - assert job completion, not just HTTP 200
APT-05	GET /api/v1/ingestion/counts	Ingestion & Scheduler (REQ-10)	GET	Unified dashboard counts read	50	5 min	TBD	TBD	TBD	TBD	P95 < 1500 ms; error rate < 1%	Not Executed	TBD	TBD	
APT-06	GET /api/v1/ingestion/status	Ingestion & Scheduler (REQ-10)	GET	Per-API ingestion status poll	20	5 min	TBD	TBD	TBD	TBD	P95 < 1000 ms; error rate < 1%	Not Executed	TBD	TBD	
APT-07	POST /api/v1/dashboard/housing/benlist	Housing Scheme - THMS (REQ-03)	POST	Beneficiary list fetch	30	5 min	TBD	TBD	TBD	TBD	P95 < 2000 ms; error rate < 1%	Not Executed	TBD	TBD	
APT-08	POST /api/v1/dashboard/patrol/camera-status	CCTV Surveillance - Patrol360 (REQ-06)	POST	Camera status snapshot	30	5 min	TBD	TBD	TBD	TBD	P95 < 1500 ms; error rate < 1%	Not Executed	TBD	TBD	
APT-09	GET /api/v1/tender/summary	Engineering Tenders - TIPS (REQ-02)	GET	Tender summary load	40	5 min	TBD	TBD	TBD	TBD	P95 < 1500 ms; error rate < 1%	Not Executed	TBD	TBD	
APT-10	GET /api/v1/housing/overall	Housing Scheme - THMS (REQ-03)	GET	Housing overall metrics	40	5 min	TBD	TBD	TBD	TBD	P95 < 1500 ms; error rate < 1%	Not Executed	TBD	TBD	
APT-11	GET /api/v1/schemes/teelp	Welfare Schemes & TELP (REQ-05, REQ-08)	GET	TELP loan summary	30	5 min	TBD	TBD	TBD	TBD	P95 < 1500 ms; error rate < 1%	Not Executed	TBD	TBD	
APT-12	POST /api/v1/ai/chat	AI Copilot Widget (REQ-20)	POST	RAG chat query	20	5 min	TBD	TBD	TBD	TBD	P95 < 5000 ms; error rate < 2%	Not Executed	TBD	TBD	LLM-backed - budget includes model inference time
APT-13	GET /api/v1/users	User Master (REQ-23)	GET	User list under load	20	5 min	TBD	TBD	TBD	TBD	P95 < 1000 ms; error rate < 1%	Not Executed	TBD	TBD	

Test ID	API Endpoint	Module	HTTP Method	Test Scenario	Concurrent Users	Duration	Avg Resp Time (ms)	P95 (ms)	Throughput (req/s)	Error Rate (%)	Target SLA	Result	Tester	Date	Remarks
APT-14	GET /api/v1/reports/te nder.pdf	Report Export (REQ-18)	GET	PDF report generation	10	5 min	TBD	TBD	TBD	TBD	P95 < 5000 ms; error rate < 1%	Not Executed	TBD	TBD	CPU-heavy rendering - consider a smaller VU target

6. Module API Integration (15 rows)

Check ID	Module	Type	Endpoint	Request Payload	Expected Result	Status	Verified By	Date	Remarks
MAI-01	TELP	COUNT	https://qatelp.pixous.info/api/Report/DistrictWise_ApplicationSummary	{ <code>"fromYear":2026,"toYear":2027,"schemelds":[""],"districtlds":["Chennai"]</code> }	District-wise application summary counts	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	Server reachable (200) but stored proc 'application_districtwise_summary' is missing in the QA DB (server-side); SSL cert expired
MAI-02	TELP	Detail	https://qatelp.pixous.info/api/Report/DistrictWise_ApplicationDetail	{ <code>"fromYear":2026,"toYear":2027,"district":"Chennai","categoryType":"statusSavedCount","skip":0,"take":0}</code> }	Application detail rows for the district	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	Same stored-proc failure as MAI-01 (server-side DB issue)
MAI-03	Tahdco Scheme	COUNT	https://scst.pixous.info/Report/GetDistrictSummary	{ <code>"fromYear":2026,"toYear":2027}</code> }	District-wise application summary	Passed	Probe (check_module_api.s.py)	11-Aug-2026	Works via GET (POST returns 405); returns district rows e.g. Ariyalur: 65 applications
MAI-04	Tahdco Scheme	Detail	https://scst.pixous.info/Report/GetApplicationDetails	{ <code>"draw":1,"start":0,"length":10,"search":{"value":"","reportFilterModel":{"districtId":"207","statusFilter":{"totalApplications":}}</code> }	Application detail rows for the district	Failed	Probe (check_module_api.s.py)	11-Aug-2026	HTTP 500 'Something went wrong'
MAI-05	TIME+Patrol 360	COUNT	https://timeqa.pixous.info/api/Report/OneDashboard_Work_Get	{ <code>"divisionIds":[],"division":["Chennai"],"district":[],"year":["2023","2024","2025","2026"],"camerastatusList":["Live","type":"work"]</code> }	Work list rows for the division	Passed	Probe (check_module_api.s.py)	11-Aug-2026	200 SUCCESS (1635 rows); SSL cert expired but backend's 'external' HttpClient bypasses it - drives the live TIPS/TIME detail flow
MAI-06	TIME+Patrol 360	Detail	https://timeqa.pixous.info/api/Report/OneDashboard_Work_Get	{ <code>"divisionIds":[],"division":[],"district":[],"year":["2026"],"camerastatusList":["Live","type":"work"],"statusNameList":[]}</code> }	Detail rows (year-filtered)	Passed	Probe (check_module_api.s.py)	11-Aug-2026	200 SUCCESS - verified end-to-end in the live dashboard (Chengalpattu/Dindigul detail grids)
MAI-07	THMS	COUNT	https://thms.tahdco.com/api/one dashboard/count	{ <code>"division":["Chennai"],"district":[],"phase":[],"terrain":["builder"]</code> }	Housing construction counts	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	403 Cloudflare Error 1010 (browser-signature/IP block from this host); works from the app itself
MAI-08	THMS	Detail	https://thms.tahdco.com/api/one dashboard/count-ben	{ <code>"division":["Chennai"],"district":[],"phase":[],"terrain":["builder"]</code> }	Beneficiary detail rows	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	Same Cloudflare Error 1010 block as MAI-07
MAI-09	TAMS	COUNT	https://tams.tahdco.com/api/one dashboard/count	{ <code>"division":["Chennai"],"district":[],"institute":[]}</code> }	Attendance counts	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	403 Cloudflare Error 1010 (browser-signature/IP block from this host)
MAI-10	TAMS	Detail	https://tams.tahdco.com/api/one dashboard/count-ben	{ <code>"division":["Chennai"],"district":[],"institute":[]}</code> }	Beneficiary detail rows	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	Same Cloudflare Error 1010 block as MAI-09
MAI-11	One Portal (Member)	COUNT	https://testncwwbv2-qa.pixoustech.app/api/OneDashboard/General	Type=MEMBER&Mode=Count&Status=HqPending&Year=2026 (query string)	Member count for the status/year	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	HTTP 200 wrapper but status=ERROR - QA DB auth failure (MYSQL5049.site4now.net) on the server side
MAI-12	One Portal (Member)	Detail	https://testncwwbv2-qa.pixoustech.app/api/OneDashboard/General	Type=MEMBER&Mode=LIST&Status=HqPending&Year=2026 (query string)	Member detail rows	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	Same QA DB auth failure as MAI-11
MAI-13	One Portal (Scheme)	COUNT	https://testncwwbv2-qa.pixoustech.app/api/OneDashboard/General	Type=Scheme&Mode=Count&Status=HqPending&Year=2026 (query string)	Scheme count for the status/year	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	Same QA DB auth failure as MAI-11
MAI-14	One Portal (Scheme)	Detail	https://testncwwbv2-qa.pixoustech.app/api/OneDashboard/General	Type=Scheme&Mode=LIST&Status=HqPending&Year=2026 (query string)	Scheme detail rows	Blocked	Probe (check_module_api.s.py)	11-Aug-2026	Same QA DB auth failure as MAI-11
MAI-15	TOD	COUNT / Detail	(not supplied)	-	Online Diary counts/details	Not Executed	-	11-Aug-2026	No endpoint provided for this module